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$$\begin{aligned} & \int \frac{dx}{x^2 - 2x + 10} \\ &= \int \frac{dx}{x^2 - 1 + 9} \\ &= \frac{1}{9} \int \frac{dx}{\frac{1}{9}(x-1)^2 + 1} \\ &= \frac{1}{9} \int \frac{dx}{\left(\frac{1}{3}(x-1)\right)^2 + 1} \\ \text{Stel } u &= \frac{1}{3}(x-1) \Rightarrow du = \frac{dx}{3} \Leftrightarrow dx = 3du \\ &= \frac{1}{3} \int \frac{du}{u^2 + 1} \\ &= \frac{1}{3} \operatorname{Bgtan}(u) + c \\ &= \frac{1}{3} \operatorname{Bgtan}\left(\frac{1}{3}(x-1)\right) + c \end{aligned}$$

References